A complete bipartite graph whose partitions have size `n` and `m` has exactly `n * m` edges.

The generic **embedding** (not necessarily planar) is defined similarly to what we have done in the previous section: it's an isomorphism Γ between a graph `G` and a subset G'=(`V`,`E`) of $\mathbb{R}^2$, such that

1. `V` is a subset of $\mathbb{R}^2$
2. Every edge `e` $\in$ `E` is a section of a *Jordan curve*[5] between two vertices
3. No two edges have the same pair of endpoints
4. No edge intersects a vertex (other than its endpoints)

Basically, with respect to the definition of plane graph and planar embedding given in section 15.1.1, we only waive the requirement that no edges can ever cross.

15.2 Planar Graphs

Kuratowski's theorem might seem counterintuitive, defining a planar graph in terms of what it can't contain. It was, however, an important tool to:

1. Recognize that two categories of graphs (complete and complete bipartite) are non-planar (except for their smallest specimens): no need to try to find ways to avoid intersections drawing them;
2. Mathematically prove when a graph isn't planar.